

Full-Stack Coding Assignment – Team Task Manager

Overview

You are required to design and develop a Team Task Management Web Application.

This assignment evaluates your full-stack development skills including frontend, backend, database design, authentication, and deployment.

The goal is to build a real-world collaborative application where multiple users can manage tasks efficiently.

Problem Statement

Build a web application where users can create and join projects, assign tasks, and track progress.

Each user should have a role (Admin or Member), and tasks should be manageable within teams.

Think of it as a simplified version of tools like Trello or Asana.

Functional Requirements

1. User Authentication

- Signup with Name, Email, Password
- Secure login (JWT or session-based)

2. Project Management

- Create projects (creator becomes Admin)
- Admin can add/remove members
- Members can view assigned projects

3. Task Management

- Create tasks (Title, Description, Due Date, Priority)
- Assign tasks to users
- Update status (To Do, In Progress, Done)

4. Dashboard

- Total tasks
- Tasks by status
- Tasks per user
- Overdue tasks

5. Role-Based Access

Admin:

- Manage tasks and users

Member:

- View and update assigned tasks only

Backend & Database Requirements

- RESTful APIs
- Database (SQL or NoSQL)
- Proper relationships (Users, Projects, Tasks)
- Validations and error handling

Deployment (Mandatory)

Deploy your full-stack application using Railway.

- Application must be publicly accessible
- Backend and frontend must be connected
- Use environment variables properly

Submission Requirements

- Live application URL
- GitHub repository

- README with setup and deployment steps
- 2–5 minute demo video explaining your project

Time Expectation

Estimated effort: 8–12 hours

Recommended timeline: 1–2 days

Important Notes

- Deployment is mandatory
- Non-working applications will not be evaluated
- Candidates must be able to explain their code

